



Assignment of master's thesis

Title:	Design and Implementation of a Scalable, On-Premise Infrastructure for Large-Scale Data Gathering, Processing, and Storage in Machine Learning Applications
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Study program:	Informatics
Branch / specialization:	Computer Systems and Networks
Department:	Department of Computer Systems
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Instructions

Machine learning (ML) applications often require large datasets, including small binary files such as images, to be efficiently stored, retrieved, and processed. In many cases, the results of ML models—such as transformed images—must be persistently stored to enable future re-evaluation, reproducibility, and iterative improvements of the models. While public cloud-based storage solutions provide such capabilities, they may not be suitable for organizations requiring full control over their data due to privacy concerns, cost considerations, or regulatory requirements.

1. Analyze the storage and processing requirements for ML-driven evaluation of small binary file datasets, specifically for image files ranging from ~10KiB to ~1MiB.
2. Research fault-tolerant distributed file systems (DFS) and evaluate open-source solutions that meet the specified requirements.
3. Choose and deploy a scalable, fault-tolerant, distributed open-source storage infrastructure optimized for small binary file storage.
4. Develop an on-demand job execution pipeline for data retrieval, storage, and ML operations on the stored data.
5. Deploy necessary infrastructure to support and orchestrate the pipeline.